# OPERATION & MAINTENANCE MANUAL

140.40 kWp Grid-Connected Rooftop Solar PV Installation

**Asset Owner:** Al Noor Packaging LLC

**Location:** Al Quoz Industrial Area, Dubai, UAE

**Primary Core Equipment:** Huawei SUN2000-110KTL-M2 & LONGi 585W HPBC

**Document Control Code:** AN-PV-OMM-2026-REV1

**Target Audience:** Internal Operations & Maintenance Department

**Date of Issue:** August 2026

# 1. Executive Summary & Environmental Operational Mandate

This dedicated document establishes the standard operating procedures, prescriptive schedule limits, and emergency safety intervention guidelines compiled specifically for the Operations and Maintenance (O&M;) Department at the Al Noor Packaging Factory in Dubai, UAE. Operating a high-performance commercial plant in an arid, high-temperature desert zone requires structured, periodic protocols to preserve the calibrated Performance Ratio (PR) of 82.3% and safely protect system investments across a projected 25-year structural lifetime.

Unmanaged solar arrays in the Al Quoz industrial cluster face severe localized soiling penalties, with dust accumulation capable of degrading day-to-day energy performance metrics by up to 2.0% per week. Adherence to the schedules within this handbook is mandatory to maintain compliance under grid net-metering conditions.

# 2. Prescriptive Cleaning & Soiling Mitigation Schedule

**Frequency Protocol:** Bi-Weekly (Every 14 Days) from May to September; Monthly from October to April.

**Execution Time Window Boundaries:** Cleaning runs must take place exclusively outside peak daylight paths (between 05:00 AM and 08:30 AM or between 06:00 PM and 08:30 PM). *CRITICAL FIELD DIRECTIVE:* Washing dark crystalline modules when internal silicon cell cores are baked under peak sunlight (reaching 75°C structural rooftop radiant heat) causes immediate thermal shock, resulting in catastrophic micro-fractures across the tempered glass face plate, instantly voiding manufacturer performance warranties.

**Mechanical Quality Criteria:** Workforces must deploy soft-bristle squeegees, specialized microfiber washing mops, and low-pressure deionized pure water streams (filtered to < 50 ppm total dissolved solids). The use of hard river water, harsh industrial detergents, or abrasive scouring pads is strictly forbidden as it etches the chemical Anti-Reflective Coating (ARC) layer of the LONGi modules, resulting in permanent, unrecoverable optical transmission drops.

**Structural Access Rules:** Technicians must navigate rows strictly utilizing the dedicated 0.50m / 0.60m internal clear walkway access corridors designed into the L1 landscape frames. Walking directly over the module glass or leaning body weights onto aluminum frame corners is strictly prohibited.

# 3. Preventive Electrical & Infrastructure Inspection Schedule

## A. Semi-Annual Infrared (IR) Thermographic Scanning (Every 6 Months)

- Perform complete thermal drone scans or handheld infrared camera checks across all 240 LONGi modules under full-load conditions (>600 W/m² irradiance). Highlight and map high-temperature hotspots crossing a $\Delta T > 10°C$ threshold. These signal bypass diode failure paths, internal ribbon delamination, or micro-cracked cells requiring swift component swaps.

### B. Quarterly Inverter Platform Auditing (Every 3 Months)

- Clean the integrated cooling fan assemblies and clear the external aluminum heat-sink fins of the Huawei 110 kW string inverter. Desert dust buildup blocks heat dissipation paths, forcing internal components to cross critical thermal limits and triggering automatic power generation throttling.
- Inspect the bottom quick-connect Amphenol MC4 ports (PV1 through PV12). Verify torque parameters read tightly and check for any signs of tracking, moisture ingress, or polymer degradation. Confirm ports PV13 through PV20 remain securely blanked off.

### C. Continuous Transient Overvoltage Monitoring

- Perform visual inspection sweeps across the mechanical health windows of the standalone DC Surge Protective Device boxes (housing 12 Phoenix Contact VAL-MS T1/T2 surge elements) and the integrated AC SPDs. A green indicator denotes normal, active safety headroom. A red mechanical flag signals that the metal-oxide varistor cartridge has absorbed a lightning strike or grid-side switching surge and must be replaced immediately.

### D. Annual Structural & Torquing Audits (Every 12 Months)

- Check mechanical connections on all click-fit racking rails and anchor systems. Re-torque every structural connection to manufacturer specifications to confirm that high wind design pressures (1,197 N/m²) cannot loosen or vibrate the array loose over time.

## 4. Technical O&M; Troubleshooting & Response Protocol

When operational telemetry alarms trigger via the active Teltonika internet gateway, field engineers must deploy the following diagnostic sequences:

| Observed Fault Symptom | Probable Engineering Root Cause | Prescribed Immediate Field Action |
|---|---|---|
| **Inverter Offline / Fault Code: Grid Loss** | The public distribution grid has shut down, or local parameters have drifted past the DEWA IPS relay windows (ANSI 27/59 loops). | Verify phase presence on the incoming busbars of the main 630A MDB. Monitor SmartLogger logs to confirm if the voltage is outside limits. Wait for automated grid synchronization timeout before force restarting. |
| **Drastic Drop in a Single MPPT Input Current** | A string branch line has developed a localized ground fault, or bypass diodes inside a panel have shorted. | Initiate an automated high-resolution **I-V Curve Scan** through the Huawei SmartLogger utility portal. Identify the exact faulty string channel (PV1-PV12), isolate the matching rooftop DC switch, and locate the physical broken panel manually. |
| **Zero-Export Fault Alarm / Total Generation Throttle** | The Modbus-RTU RS485 communication link between the smart kWh meter and the SmartLogger 3000A has split or encountered high noise. | Perform a continuity test on the shielded, twisted-pair signaling lines running down to the 630A incomer panel. Verify the three CT sensing rings remain clamped and powered. Re-establish communication links to restore full solar inverter tracking. |

## 5. Main Earthing & Grounding Integrity Verifications

To guarantee high-voltage lightning path diversion and preserve safety compliance targets, the O&M; department must execute yearly earth impedance loop tests:

- **Ground Loop Resistance Checking:** Isolate each individual earth pit node and measure the systemic impedance path using a calibrated earth ground tester. The cumulative path path resistance must pass utility inspector sign-off with a rigid value of $\mathbf{R \leq 1.0\ \Omega}$. If values drift higher, treatment of the surrounding soil matrix with grounding enhancement compounds is mandatory.
- **Equipotential Bonding Proofing:** Verify that the 16 $mm^2$ yellow/green insulated copper core wire running from the inverter frame chassis down to the primary building earth busbar maintains tight, corrosion-free contact.

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